Muhammad Ali Khan

Pécs, Hungary | muhammadalikhanhitec@gmail.com | +36 20 578 8983 | Portfolio | LinkedIn | GitHub

EDUCATION

University of Pécs — Pécs, Hungary
BSc Computer Science — *Sep. 2024 to Jul. 2027*

Aalto University Summer School — Helsinki, Finland
Introduction to Artificial Intelligence and Machine Learning — *Jul. 2025 to Aug. 2025*

EXPERIENCE

Research Intern — 29 May 2026 to 31 Jul. 2026
Trustworthy Artificial Intelligence Laboratory (TAIL), University of Cagliari — *Cagliari, Italy*

Used FlashRAG to build and evaluate retrieval pipelines, datasets, and benchmarks for RAG research.
Used FastAPI and Docker to run Mosaic RAG components separately during information retrieval and LLM experiments.

PROJECTS

Traffic and Crowd Monitoring System [GitHub]
Current BSc thesis that uses Python, OpenCV, YOLO, and PostgreSQL to detect objects in aerial images and video, store results, and evaluate the baseline.

Multi UAV Mission Planning [GitHub]
Developing a MATLAB simulation that uses UAV Toolbox, terrain, and OpenStreetMap building data for multi UAV delivery missions.

Smart Sonar Surveillance System [GitHub]
Built a Raspberry Pi prototype with an ultrasonic sensor and servo to scan 0 to 180 degrees, measure distance, and trigger proximity alerts.

Mosaic RAG [GitHub]
Built a retrieval augmented generation system with FastAPI and Docker to separate retrieval and generation steps for LLM experiments.

TECHNICAL SKILLS

Programming: Python, C++, Java, SQL, TypeScript
AI / Computer Vision: PyTorch, OpenCV, YOLOv8, FlashRAG
Backend / DevOps: FastAPI, Docker, PostgreSQL, Git, GitHub Actions
Engineering Tools: MATLAB, Simulink, UAV Toolbox, Autodesk Fusion, Altium Designer, OpenRocket

ACTIVITIES AND CERTIFICATIONS

WATT Saxion Association, Study Trip Committee — Sep. 2023 to Mar. 2024
Served on the Study Trip Committee.

NASA International Space Apps Challenge — Oct. 2024
Built a team web application on geomagnetic storms and their effects on power infrastructure.

Google Cybersecurity Professional Certificate — Jul. 2025
Completed eight courses covering Linux, SQL, Python, SIEM, intrusion detection, and risk assessment.

C++ Programming Certificate, ACE — May 2023
Completed a two-month C++ programming course with a B grade.

AWS Certified Solutions Architect - Associate — In progress
Preparing for the associate-level cloud architecture certification.

INTERESTS

Interests: Systems Engineering, Systems Programming, Autonomous Systems, Embedded Systems
Systems and DevOps: Linux, Docker, Git, GitHub Actions, FastAPI, PostgreSQL